

Fig. 2: An actual-size, single-side PCB for device for selecting diodes

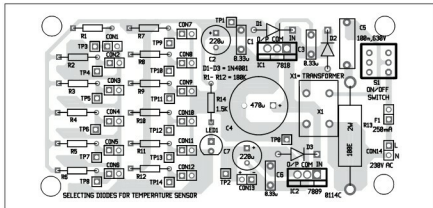


Fig. 3: Component layout of the PCB

for a diode will be:

$$TKU = (U1-U2)/(T1-T2), \text{ mV}/^{\circ}\text{C}$$

The voltage drop change of the PN junction due to temperature change is linear, so we should take measurements at only two temperatures for each diode. The test temperatures can be the same for all the diodes

or different. Most of the diodes can be used in the temperature range of -25°C to +150°C without any problem. The temperature coefficient of most of the PN junctions is in the range of -1.3mV/°C to -3mV/°C, so we can use these diodes as sensors.

Construction and testing

An actual-size, single-side PCB for the device is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable plastic case.

To test the circuit for proper functioning, verify correct power supply for the diodes and DVM at TP1 and TP2 with respect to TP0. Voltage drop across each diode can be verified at TP3 through TP14. ●

The author worked as a researcher and assistant professor in Technical University of Sofia (Bulgaria) and a lecturer in Kingdom of Morocco before he started working as an electronics engineer in the private sector in Bulgaria

PARTS LIST

Semiconductors:

- D1-D3 - 1N4001 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1-R12 - 180-kilo-ohm
- R13 - 100-ohm, 2W
- R14 - 1.5-kilo-ohm

Capacitors:

- C1, C3, C6 - 0.33µF ceramic disk
- C2 - 220µF, 35V electrolytic
- C4 - 470µF, 40V electrolytic
- C5 - 0.1µF, 630V ceramic disk
- C7 - 220µF, 16V electrolytic

Miscellaneous:

- CON1-CON13 - 2-pin connector
- DVM - For digital multimeter
- X1 - 230V AC primary to 21V, 250mA secondary transformer

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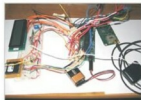
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